rfPhen2Gen: A machine learning based association study of brain imaging phenotypes to genotypes

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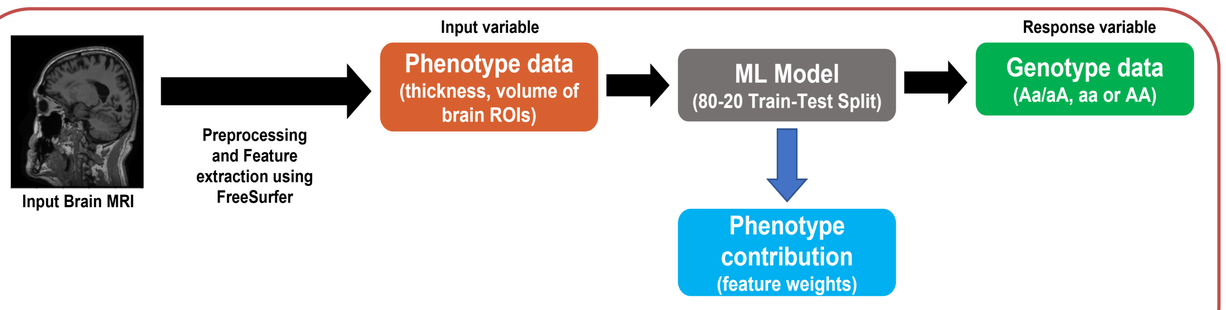
INTRODUCTION

- Imaging genetic studies aim to find associations between genetic variants and imaging quantitative traits [1].
- Traditional genome-wide association studies (GWAS) are based on univariate statistical tests [2].
- However, when multiple traits are analyzed together, they suffer from a multiple-testing problem and from not taking into account correlations among the traits [3].

Standard GWAS approaches vs 'Reverse' GWAS approach

METHODS

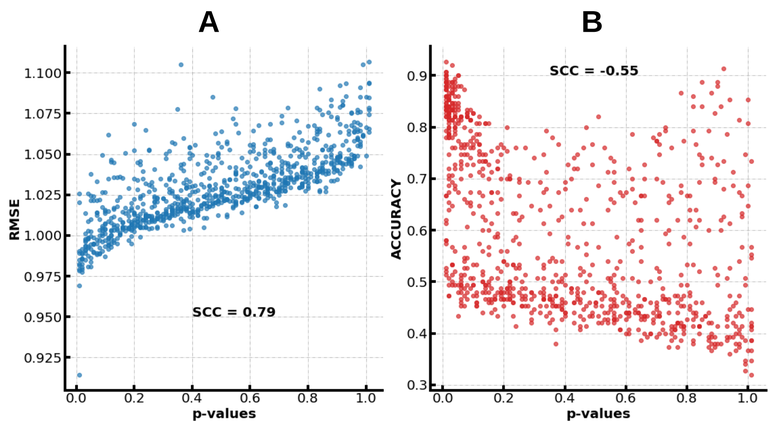
- An alternative approach to multi-trait GWAS is to reverse the functional relation between genotypes and traits [4].
- However, current reverse genotype prediction approaches are mostly based on linear models.
- Here, we evaluated random forest regression (RFR) as a method to predict 518,484 SNPs from 56 brain imaging QTs (obtained from ADNI database [5]) and identify biologically relevant associations.



Flowchart of the approach used in this study

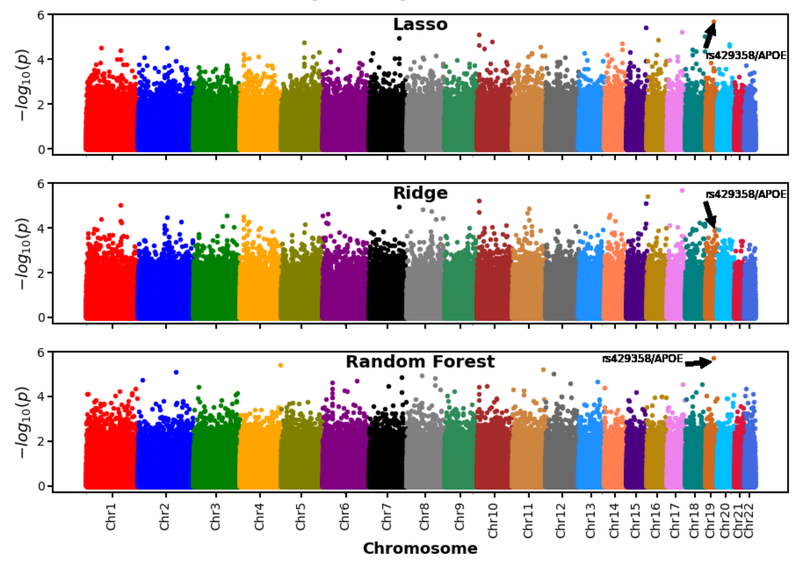
RESULTS

1. Genotype regression error is a <u>better indicator</u> of permutation *p-value* significance than genotype classification accuracy.



Permutation tests for Random Forests. A. Regression (RMSE), B. Classification (Accuracy). *SCC: Spearman Correlation Coefficient*

2. SNPs within the known Alzheimer disease (AD) risk gene <u>APOE</u> had lowest RMSE for *lasso* and *random forest*, but not *ridge regression*.



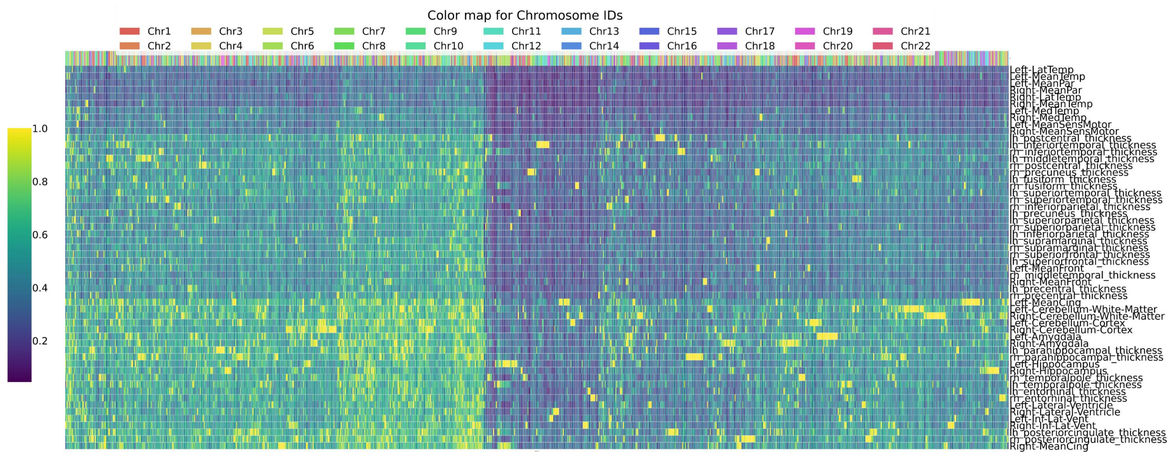
Negative log-transformed *p-values* plotted across the whole genome for Random Forest Regression, Ridge Regression, Lasso Regression

3. Feature selection identified well-known brain regions associated with brain-related disorders (e.g., *hippocampus*, *amygdala* etc.) as important predictors of the most significant SNPs

SNP	Gene	Associated traits	Features identified
rs429358	*APOE*	HDL/LDL cholesterol, Fatty liver, Alzheimer's disease	Hippocampus, Amygdala, Entorhinal cortex
rs2084729	*F11-AS1*	cognitive decline, cortical thickness/surface area, brain volume measurements	Entorhinal cortex, Amygdala, Posterior cingulate cortex
rs622735	*FLI1*	PHF-tau measurements	Parahippocampal gyrus, Superior Temporal gyrus
rs3749030	*TTC21B*	intelligence, cognitive function	MeanCing, Cerebellum White Matter, Postcentral gyrus
rs11048593	*ITPR2*	HDL/LDL/Total cholesterol, unipolar depression	Cerebellum White Matter, MeanCing, Amygdala
rs6473635	*PXDNL*	cognitive impairment measurement, unipolar depression, bipolar disorder, schizophrenia	MeanCing, Temporal pole gyrus, Amygdala
rs896189	*TMEM213*	PHF-tau measurements	Inferior parietal gyrus, postcentral gyrus, fusiform gyrus
rs12678956	*SYBU*	Schizophrenia, cortical surface area	Inferior parietal gyrus, posterior cingulate cortex, fusiform gyrus
rs6714955	*GACAT3*	cortical surface area, neuroimaging measurements, brain measurements	Posterior cingulate cortex, Cerebellum cortex, MeanCing
rs9482965	*LAMA2*	cognitive ability, cortical thickness/surface area	MeanTemp. posterior cingulate cortex, Inferior lateral ventricle

Top 10 SNPs identified by random forest

4. Clustering of image features show that groups of variants tend to associate with similar brain regions or features



Heat maps of feature weights for top 1000 SNPs identified by random forest

Conclusion

- Our results indicate that non-linear methods like random forests may offer additional insights into phenotype-genotype associations compared to traditional linear multi-variate GWAS methods.

References

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